MIL-C-915A(SHIPS)
AMENDMENT - 8
13 March 1959
SUPERSEDING
Amendment - 7
9 June 1958

INTERDEPARTMENTAL SPECIFICATION

CABLE, CORD AND WIRE, ELECTRICAL (SHIPBOARD USE)

This amendment forms a part of the Military Specification MIL-C-915A(SHIPS), 30 July 1952.

Page 1, paragraph 1.2.1.1: Delete reference to type SHFA, DHFA, FHFA, THFA, SHFP and HA". (Note. - Types SSGA, DSGA, TSGA, FSGA, SS5P and MSCA of Specification MIL-C-2194

[redacted] and PBTX: Delete "lead wire" and substitute "lead,

"Type PBTM - Pyrometer base, multiple pairs, armored."

Page 1, paragraph 1.2.1.2: Add:

"Type SSP - Single, special purpose, flexible."

Page 2, paragraph 1.2.2.1: Add:

"Type T3P - Triple-conductor, 3000 volt, portable."

Page 2, paragraph 1.2.2.2: Add:

"Type DLT - Divers' lifeline and telphone cable.
"Type JAS - Jet aircraft servicing."

Page 2, paragraphs 1.2.3 and 1.2.3.1: Delete. (Note. - Type FHOF-4 replaces this type.)

[redacted]paragraph 1.2.4: Delete reference to "types SRI and SRIB". (Note. - Types B, C and D of [redacted] replace these types.)

[redacted] Military Specifications: Add:

"MIL-C-5423[redacted]osion-Resistant), Flexible, Preformed (for Aerona[redacted])."

Pages 2 and 3, paragraph 2.1. under Navy Department: Delete reference to "2264".

Pages 2 and 3: Add the following [redacted] Drawing:

"DRAWINGS"

FSC 6145

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Page 3: Add paragraph 2.2:

"2.2 Other publications. - The following document form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS
D27-54T:

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

NATIONAL BUREAU OF STANDARDS
Circular No. 561.

(Application for copies should be addressed to the National Bureau of Standards, [redacted]

Page 3, paragraph 3.1.1: Delete and substitute:

"3.1.1 Electrical cable, except types TRF, TRXF, DBSP, TBSP, FBSP, MCGC, MCSC, DCOP, TCOP, SHOF, MRI, SHFS and SSF, furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.3)."

Page 4, paragraph 3.2.17, last line: Delete and substitute "cable in coils (see 5.1.1.1) is 500 feet unless otherwise stated on the individual specification sheet".

Page 5: Add as paragraph 3.2.22:

"3.2.22 Manufactured length. - A manufactured length shall consist of any one single length of cable to which the impervious sheath was applied in one continuous operation without stoppage or change in machine set-up."

Page 5, paragraph 3.6, line 2: Delete "(see 3.12 to 3.25, inclusive, 3.27, 3.28 and 3.34)" and substitute "(see 3.19 to 3.22 inclusive, 3.24 to 3.28 inclusive, 3.32, 3.34 and 3.46)".

Page 6, paragraph 3.8.2: In line 1, delete "approximately 1/8 inch" and substitute [redacted] 1/10"; in line 2, delete "cotton or cellulose" and substitute "cotton, cellulose [redacted]

Page 7, paragraph 3.9.5: Add "Where two tracers [redacted] directions."

Page 10, table III, column 1, under "Bunch:", line 28: Delete "3/4(7)" and substitute "3/5(7)".

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Page 10, table III, under "Rope", line 2, columns 5 and 6: Delete "22,190 and 22,910" and substitute "22,910 and 22,460" and insert the following:

Standard copper conductor size	Number of strands	Strand diameter	Diameter over conductor	Cross sectional area of conductor		Maximum conductor resistance (d.c.) per 1000 feet at 25°C.		Weight per 1000 feet
	Minimum	Nominal	Nominal	Nominal	Minimum	Bare	Coated	Approximate
		Inch	Inches	Cir. Mils	Cir. Mils	Ohms	Ohms	Pounds
"250 (6384)	19x7x48	.008	.713	250,000	245,000	.0461	---	821
300 (259)	37x7	.034	.714	300,000	294,000	.0378	---	926"

Page 11, table IV: Add:

Standard copper wire sizes					Maximum resistance (d.c.) per 1000 feet at 25°C.		Approximate weight per 1000 feet
A.W.G. No.	Strand diameter		Cross sectional area				
	Nominal	Calculated	Nominal		Bare	Coated	
	Inch	Inch	Cir. Mils		Ohms	Ohms	Pounds
"---------	.034	.03403	1158		9.60	9.79	3.51"

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Page 11, paragraph 3.10.4.1, line 4: Delete "of lay or" and substitute "of lay of".

Page 12, paragraph 3.10.6: Add "Iron and constantan pyrometer base lead wire may be welded as necessary to obtain the specified lengths."

Page 18, paragraph 3.11.10.4, line 3: Delete "3.11.6" and substitute "3.11.16".

Page 19, paragraph 3.11.11.2.3, line 2: Delete "silver" and substitute "sliver".

Page 19, paragraph 3.11.11.3, lines 1 and 2: Delete "1.2.2 and 1.2.3.1" and substitute "and 1.2.2".

Page 23, paragraphs 3.11.20.2 and 3.11.20.3: Delete.

Page 26, paragraph 3.12: Delete and substitute:

"3.12 Type SSF cable. -

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see note 1).
Second. A separator (see 3.11.14).
Third. A wall of synthetic rubber (see 3.11.3) or butyl rubber (see 3.11.5) having a minimum wall thickness (see 3.7.3) of 0.075 inch.
Fourth. A reinforcement (see 3.11.12) at manufacturer's option.
Fifth. A polychloroprene sheath (see 3.11.17) bonded to the underlying insulation, having a minimum wall thickness (see 3.7.3) of 0.075 inch.

LENGTH REQUIREMENT

Shipping lengths (feet) - 1045 minimum, 1155 maximum.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Cold bend (see 4.8.9). No cracking of sheath when bent two turns around mandrel with a diameter twice that of specimen at -30°C.
Electrical tests (see following table).

Type	Standard copper conductor	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see note 1)	Insulation resistance per 1000 feet (see 4.8.25)
		Min.	Max.	Min.	Max.	Min.
		Inches	Inches	Volts	Ohm	Megohms
SSF-300	300 (259)	1.020	1.100	2500	0.0388	100

Note 1. - The copper shall be hard drawn with a minimum tensile strength of 67,200 p.s.i. and a minimum elongation of 0.85 before stranding. The resistance specified is based on 102.7 percent of the values required for annealed copper."

Pages 27 through 32, paragraphs 3.13 through 3.18: Delete.

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Page 33, paragraph 3.19: Delete and substitute:

"3.19 Type SHFR, DHFR and THFR cable (for 3,000 volt service). -

CONSTRUCTION

First.	A standard conductor of bare, uncoated copper (see 3.10).	Seventh.	Conductors of DHFR and THFR shall be twisted together with a lay not greater than 24 times the pitch diameter. Fillers (see 3.11.11) shall be employed to obtain a firm, well rounded assembly.
Second.	A wall of synthetic resin insulation (see 3.11.2).	Eighth.	A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Third.	With or without separator (see 3.11.14) at manufacturer's option.	Ninth.	With or without separator (see 3.11.14) at manufacturer's option.
Fourth.	A wall of varnished-cambric insulation (see 3.11.7).	Tenth.	An impervious sheath (see 3.11.18).
Fifth.	A separator (see 3.11.14).	Eleventh.	Braided metal armor (see 3.11.20).
Sixth.	A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10), with or without separator (see 3.11.14) on SHFR at manufacturer's option.	Twelfth.	Paint (see 3.11.21).

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Standard copper conductor	Number of conductors	Diameters: Over synthetic resin, Minimum	Diameters: Over varnished cambric	Diameters: Over first asbestos or glass	Diameters: Overall	Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare, Minimum	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare, Maximum	Insulation resistance 1000 feet (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9)
			Inch	Inch	Inch	Inch	Volts	Ohms		Megohms	Inches
SHFR-4	4(7)	1	0.115	0.185	0.247	0.540	8000	2.46		300	7
DHFR-4	4(7)	2	.115	.185	.247	.844	8000	2.50		400	10
THFR-4	4(7)	3	.115	.185	.247	.883	8000	2.50		400	11

Page 35, paragraph 3.21, item "Eighth", line 5: After "19(6)" insert "and 19(14)".

Page 36, paragraph 3.22, item "Eighth", line 4: Delete "shall" and substitute "may" and line 5, delete "in size 19(14)"

Page 37, paragraph 3.23: Delete.

Page 38, paragraph 3.24: Delete "Sixth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10)."

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Page 39, paragraph 3.25, item "First": Add "Temperature electromotive force relationships shall follow National Bureau of Standards Circular No. 561."

Page 40, paragraph 3.26: Delete and substitute:

"3.26 Types DSS, TSS, FSS and MSS. -

CONSTRUCTION

First. A standard copper conductor with metallic coating and watertight compounds (see 3.6 and 3.10).

Second. A wall of synthetic insulation (see 3.11.3 or 3.11.5).

Third. (a) Types DSS, TSS and FSS shall have 2, 3 and 4 insulated conductors, respectively.
(b) Type MSS-6 shall have six insulated conductors comprising two shielded pairs and two unshielded singles. Each shielded pair shall comprise two insulated conductors twisted together with a braided shield of 0.0063-inch diameter tinned copper wire over the pair. The braid angle shall be approximately 30 degrees (see 3.11.16.6).
(c) Color coding is required and may be obtained at manufacturer's option by colored insulation (see 3.9.2), or braid (see 3.9.5), or by application over the insulating wall of flexible coating film of colored lacquer or paint which will not crack, peel or fade in the completed cable. Samples of film coated wire with identification of the coating material shall be submitted to and approved by the bureau or agency concerned prior to production. Colors in 2, 3 and 4 conductor cable shall be black and white, black and white and red, black and white and red and green, respectively. The colors in type MSS-6 shall be black and white in one pair, red and blue in the other pair, one yellow single and one green single.

Fourth. The required number of conductors and shielded pairs shall be twisted together with a lay not exceeding that specified in the following table. Fillers of synthetic rubber like compound shall be employed with a minimum amount of fibrous material to obtain a firm, well-rounded cross section.

Fifth. An approved binder over the conductor assembly.

Sixth. A close braid of 0.0063-inch tinned copper wire providing at least 88 percent coverage on types DSS, TSS and FSS.

Seventh. A polychloroprene sheath (see 3.11.17). The assembled components shall be molded together compactly to enable the completed cable to meet the hydrostatic test. The individual conductors shall be readily separable from fillers and binders without damage to the insulation.

LENGTH REQUIREMENT

Nominal length (see 5.1.1.1) 500 foot coils unless otherwise specified in the contract or order.

TEST REQUIREMENTS

Electrical tests (see 4.8.19 and following table):
Dielectric strength:
(a) 1500 volts r.m.s. conductor to conductor.
(b) 1000 volts r.m.s. conductor to shield.
(c) 500 volts r.m.s. shield to water.
Insulation resistance:
(a) 200 megohms/1000 feet between conductors and conductor to shield.
(b) 10 megohms/1000 feet shield to water.
Flammability (see 4.8.10) - Ignition time, 30 seconds, minimum burning time, 120 seconds, maximum flame travel, 2 inches, maximum.

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Flexing endurance (see 4.8.14) - Except 1000 cycles of bending only on a 4 inch diameter mandrel at room temperature.
Induced voltage (see 4.8.29) - 5 microvolts maximum for type MSS-6 only.
Hydrostatic test (see 4.8.31 and Appendix I) - Maximum permissible leakage 0.1 cubic inch during 2 hours at 500 pounds per square inch. A special packing gland, conforming to Drawing 9000-S6202-1197101, obtainable from the Government inspector, shall be used for conducting this test on type SS cables.

Type	Standard copper conductor	Number of conductors	Insulation thickness (see note 1)	Lay	Sheath thickness (see note 1)	Overall diameter		Conductor resistance (d. c.) per 1000 feet at 25°C.
			Minimum	Maximum	Minimum	Minimum	Maximum	Maximum
			Inch	Inches	Inch	Inch	Inch	Ohms
DSS-2	2(7)	2	0.020	2	0.050	0.370	0.390	6.64
DSS-3	3(7)	2	.025	2-1/2	.060	.480	.500	4.15
DSS-4	4(7)	2	.025	2-1/2	.060	.480	.500	2.57
TSS-4	4(7)	3	.025	2-1/2	.050	.480	.500	2.57
FSS-2	2(7)	4	.020	2-1/2	.070	.480	.500	6.64
FSS-4	4(7)	4	.025	3.0	.070	.600	.625	2.57
MSS-6	1(7)	6	.020	3.0	.060	.600	.625	10.5

Note 1. - Because of the compacted construction necessary to meet the hydrostatic test required on these cables, the centering and circularity of the insulation and sheath (see 3.7.4) is waived provided the specified minimum wall (see 3.7.3) is met, and the least thickness at any point (see 3.7.3) is not less than 80 percent of the specified value."

Page 41, paragraph 3.27: Delete and substitute:

"3.27 Type TSP cable. -

CONSTRUCTION

First. A standard copper conductor with metallic coating, size 3/5(7) (see 3.6 and 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. Two conductors twisted together with a maximum lay of 2-1/2 inches to form a twisted pair with color coded insulation (see 3.9.7.2).
Fourth. The specified number of twisted pairs cabled together with a maximum lay of 10 inches in the outer layer. Watertight sealing compounds but no fibrous fillers shall be employed.
Fifth. A suitable binder and/or separator may be applied over the assembled pairs.
Sixth. An impervious sheath (see 3.11.18) of the thermoplastic type.

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TEST REQUIREMENTS

Flammability (see 4. 8. 10).
Electrical tests (see following tabulation).
Watertightness (see Appendix I).

Type	Number of		Thickness of		Overall diameter	Dielectric strength conductor to ground	Insulation resistance per 1000 feet	Conductor resistance per 1000 feet at 25°C.
			Insulation	Sheath				
	Conductors	Pairs	Minimum	Minimum	Maximum	Minimum	Minimum	Maximum
			Inch	Inch	Inches	Volts	Megohms	Ohms
TSP-11	22	11	0.015	0.080	0.735	2,500	100	16.6
TSP-31	62	31	.015	.080	1.062	2,500	100	16.6

"

Pages 43 and 44, paragraph 3. 29: Delete and substitute:

"3. 29 Type TTRSA cable. -

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3. 10) size 1(7).
Second. A wall of polythene (see 3. 11. 4) to the diameter and within the dimensional tolerances necessary to comply with all performance test requirements and overall diameter of the completed cable. At the manufacturer's option, a thin wall of suitable insulating material may be applied over the polythene to enable the completed cable to meet the thermoplastic flow test (see 4. 8. 32).
Third. Two such insulated conductors twisted together with a maximum lay of 2-1/2 inches to form a pair. One conductor in each pair to be white or natural and its mate shall be a dark contrasting color such as blue or black.
Fourth. A braided shield of 0. 005 or 0. 0063-inch diameter bare or tinned copper wire (see 3. 11. 16. 6) on each pair. The braid angle, shall be approximately 30 degrees.
Fifth. Insulation over shield to withstand voltage test between adjacent shields, obtained with an inner braid (see 3. 11. 16. 2) treated to prevent fraying at cut ends incident to handling, or by other means approved as a component of the qualified product. Color coding or printed numbers (see 3. 9. 5) over each insulated shield.
Sixth. The specified number of shielded pairs cabled together with a lay not exceeding 14 inches. Fillers (see 3. 11. 11) shall be employed where necessary to obtain a firm, well rounded assembly.
Seventh. A wall of felted asbestos (see 3. 11. 9) or glass fiber (see 3. 11. 10) over the assembly of pairs.
Eighth. An impervious sheath (see 3. 11. 18).
Ninth. Braided metal armor (see 3. 11. 20).
Tenth. Paint (see 3. 11. 21).

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TEST REQUIREMENTS

Cold bend (see 4.8.9 and the following table).
Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Electrical tests:
Insulation resistance (see 4.8.25) - 1000 megohms per 1000 feet.
Conductor resistance (see 4.8.24) - (see following table).
Dielectric strength (see 4.8.19) - (see following table).
Capacitance (see 4.8.27) - 28.0 micromicrofarads (maximum) per feet at 25°C. and 1 megacycle per second.
Characteristic impedance (see 4.8.27) - 75.0 ohms (minimum) and 90.0 ohms (maximum) at 25°C. and 1 megacycle per second.
Thermoplastic flow test (see 4.8.32) - no electrical failure in 6 hours.

Type	Number of shielded pairs	Overall diameter (max.)	Dielectric strength, (min.)			Conductor resistance (d.c.) per 1000 feet at 25°C. (max.)	Cold bend mandrel diameter
			Conductor to mate	Conductor to shield	Shield to shield		
		Inches	Volts	Volts	Volts	Ohms	Inches
TTRSA-2	2	0.740	2000	1000	200	10.3	9
TTRSA-4	4	.800	2000	1000	200	10.3	10
TTRSA-6	6	.940	2000	1000	200	10.3	12
TTRSA-8	8	1.050	2000	1000	200	10.3	13
TTRSA-10	10	1.140	2000	1000	200	10.3	15
TTRSA-12	12	1.160	2000	1000	200	10.3	15
TTRSA-16	16	1.250	2000	1000	200	10.3	16

Page 45, paragraph 3.30, item "First.", last sentence: Delete and substitute "The steel and stranding shall conform to Specification MIL-C-5424."

Page 46, paragraph 3.31, under "Test Requirements": Delete "Heat shock (see 4.8.30) - no cracking of insulation or sheath" and substitute "Heat Shock (see 4.8.30) - Dissection after test shall reveal no cracking of insulation or sheath, and no damage such as tears or splits in the copper tape."

Page 47, paragraph 3.32: Delete and substitute:

"3.32 Type FCSF cable. -

CONSTRUCTION

First. Standard conductors of bare, uncoated copper (see 3.6, 3.10 and the following tabulation). At manufacturer's option, a concentric strand (see 3.10.4.1) of 37 wires in lieu of 20(49) and 91 wires in lieu of 66(133) may be furnished, provided the specified conductor resistance is not exceeded.
Second. A separator (see 3.11.14).
Third. A wall of butyl rubber (see 3.11.5).
Fourth. A compound filled tape (see 3.11.15) or an inner braid (see 3.11.16.2) on each conductor is permitted but not required. Color coding, one black and one white on each conductor size is required.

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Fifth. The four conductor cabled together with fillers of synthetic rubber or polychloroprene to obtain a firm, well rounded assembly, with a minimum amount of fibrous material. Outer fillers may be integral with the sheath compound or may be laid in separately.
Sixth. A suitable binder over the conductor assembly is permitted but not required.
Seventh. A polychloroprene sheath (see 3.11.17). Fillers and sheath material in the completed cable shall be readily separable from, and without damage to, the conductor insulation. Reinforcement (see 3.11.12) shall be applied under the sheath, or under at least half the sheath thickness.

LENGTH REQUIREMENT

Shipping lengths (feet) - FCSF-66 - 1500 min., 1650 max.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Cold bend (see 4.8.9). No cracking of sheath when bent two turns around mandrel with diameter twice that of specimen at -30°C.
Electrical tests (see following table).
Watertightness (see appendix I): Apply leakage limit for larger conductor size.

Type	Standard copper conductor	Number of conductors	Insulation thickness		Lay of conductors	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 ft. at 25°C. (see 4.8.24)	Insulation resistance per 1000 ft. (see 4.8.25)
		Total	Minimum	Maximum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
			Inch	Inches	Inches	Inches	Inches	Volts	Ohm	Megohms
FCSF-66	2 of 20(49) and 2 of 65(133)	4	0.062 .062	12	1.330	1.370	2500	0.573 .174	300

Page 49, paragraph 3.34, tabulation, column 8: Delete "2000" in both places, and substitute "200"

Page 51, paragraph 3.36, tabulation: Delete reference to "type MHFF-2 and type MHFF-4" "(Note: These two types have been replaced by types DHOF-3 and FHOF-3.)"

Page 53, paragraph 3.38, under "Test Requirements", column 2, line 3: After "Five microvolts" add "(max.)"

Page 54, paragraph 3.39, under "Construction" item "Fifth": Delete and substitute:

"Fifth. An impervious sheath (see 3.11.18)."

Page 56, paragraph 3.41, under "Test Requirements": Delete "Impact resistance (see 4.8.15)" and "Abrasion resistance (see 4.8.16)."

Page 57, paragraph 3.42, under "Test Requirements": Delete "Impact resistance (see 4.8.15)" and "Abrasion resistance (see 4.8.16)."

Pages 58 and 59, paragraph 3.43: Delete and substitute:

"3.43 Type TTRS cable. -

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10), size 1(7).
Second. A wall of polythene (see 3.11.4) to the diameter, and within the dimensional tolerances required to obtain the specified degree of performance as limited by the maximum overall diameter of the finished cable.
Third. Two such insulated conductors shall be combined to form a pair. They shall be twisted together with any lay necessary to obtained the specified mechanical and electrical characteristics. One conductor in each pair shall be white or natural and its mate shall be a dark contrasting color such as blue or black.

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Fourth. A braided shield of 0.005 or 0.0063 inch diameter bore or tinned copper wire (see 3.11.16.3) on each pair. The braid angle shall be approximately 30 degrees.

Fifth. An inner braid (see 3.11 16.2) on each pair, treated with clear flame and moisture resistant compound to prevent fraying at cut ends incident to handling during installation. Color-coding (see 3.9.5) shall be provided for each shielded pair.

Sixth. The specified number of shielded pairs shall be cabled together with a left-hand lay. The length of lay shall be a suitable function of the pitch diameter of that layer, designed to meet the flexing endurance test on completed cable. Wherever practicable, the component pairs shall be cabled in a single layer, using, if necessary, a soft center core. A single pair shall not be used as a core for the cabled construction. Fillers (see 3.11.11.3) shall be used if necessary to obtain circularity of cross section.

Seventh. The cabled pairs shall be bound by a loose braid or fabric binder in such a manner as to permit sliding between the pairs themselves and between the pairs and the overall braid or binder when the cable is sharply and repeatedly bent and twisted.

Eighth. An impervious sheath (see 3.11.18) of the thermoplastic type.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14).
1000 twisting and bending cycles at 75°C.
500 twisting and bending cycles at minus 30°C.

ELECTRICAL TESTS

Conductor resistance (see 4.8.24) - 10.3 ohms (max.) per 1000 feet of cable at 25°C.
Insulation resistance (see 4.8.25) - 1000 megohms per 1000 feet (min.).
Dielectric strength (see 4.8.19) - 5 minutes, 2000 volts r.m.s. conductor to mate and 1000 volts conductor to shield (min.).
Capacitance (see 4.8.27) - 26 micromicrofarads per foot (max.) at 25°C. and 1 megacycle.
Characteristic impedance (see 4.8.27) - 75 ohms (min.), 90 ohms (max.) at 25°C. and 1 megacycle.

Type	Number of shielded pairs	Diameter over sheath
		Inches
TTRS-2	2	0.680
TTRS-4	4	.740
TTRS-6	6	.880
TTRS-8	8	.990
TTRS-10	10	1.080
TTRS-12	12	1.100
TTRS-16	16	1.190

"

Page 60, paragraph 3.44: Under "Construction", item "First", line 2, delete "(see 3.11.3.2 and 3.11.5)" and substitute "(see 3.11.3 or 3.11.5)", and in item "Third", line 3, delete "(see 3.11.3) and substitute "(see 3.11.3 or 3.11.5)".

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Page 61, paragraph 3.45: Delete and substitute:

"3.45 Type JAS cable. -

CONSTRUCTION

First. Standard conductors of bare uncoated copper (see 3.10 and following tabulation) with separator (see 3.11.14).
Second. A wall of butyl rubber (see 3.11.5).
Third. An inner braid (see 3.11.16.2) on size 6(65). A tape (see 3.11.15) or inner braid (see 3.11.16.2) on size 250(6384) is not required but will be permitted at manufacturer's option. Color coding, one conductor white and one conductor black of each size.
Fourth. Two conductors of size 6(65) twisted with a maximum lay of 4 inches to form a pair with fillers (see 3.11.11.3). Two conductors of size 250(6384) laid parallel with the twisted pair in between in the geometric center. Fillers of synthetic elastomer or polychloroprene shall be employed to fill all valley spaces and make a firm oval cross section. At manufacturer's option, fillers may be an integral part of the sheath.
Fifth. An open mesh reinforcement of type P cord in accordance with Specification MIL-S-572 over the conductor assembly.
Sixth. A polychloroprene sheath (see 3.11.17). Fillers and sheath material shall be readily removed from the completed cable without damage to the conductor insulation.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Bending endurance (see 4.8.14) - No electrical failures, broken conductors or physical damage during 1000 cycles flatwise room temperature bending only on 14-inch diameter mandrel.

ELECTRICAL TESTS

Dielectric strength - 2000 volts 5 minutes (min.).
Insulation resistance, megohms per 1000 feet (min.) - 200 for size 6(65) and 100 for size 250(6384).
Conductor resistance, ohms (d.c.) per 1000 feet at 25°C. (max.) - 176 for 6(65) and 0.0461 for 250(6384).

Type	Conductors				Insulation wall [1]	Sheath wall	Overall diameter (oval shape)
	Number of conductors	Strands per conductor	Size of strand	Conductor diameter			
		Minimum	Nominal	Nominal	Minimum	Minimum	Maximum
			Inch	Inch	Inch	Inch	Inches
JAS-250	2 of 6(65) and 2 of 250(6384)	65 6384	0.010 .006	0.097 .713	0.031 .070	0.156	1.260 minor and 2.480 major

[1] When the insulation is applied as a "D" shaped section to fit halves of the oval lay-up, the eccentricity requirement (see 3.7.4.1) is waived."

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Page 62, paragraph 3.46: Delete and substitute:

"3.46 Type PBTM cable. -

CONSTRUCTION

First. One conductor of bare copper and one conductor of constantan (see 3.6 and 3.10). Temperature electromotive force relationships shall follow National Bureau of Standards Circular No. 561. Any necessary joints shall be silver soldered in copper wire and welded in constantan wire.
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. One copper and one constantan conductor twisted with a maximum lay of 3 inches to form a pair.
Fourth. The specified number of twisted pairs cabled with a maximum lay of 10 inches in the outer layer. Nonfibrous sealing-compound fillers in all valley spaces (see 3.6). Telephone circuit identification (see 3.9.7.2).
Fifth. An approved binder over the assembled pairs to form a round core.
Sixth. An impervious sheath (see 3.11.18).
Seventh. Braided metal armor (see 3.11.2) and paint (see 3.11.21).

TEST REQUIREMENTS

Cold bend (see 4.8.9)
Flammability (see 4.8.10)
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Conductors			Diameter over insulation	Diameter, overall	Dielectric strength	Conductor resistance (d.c.) per 1000 feet at 70°F.	Insulation resistance per 1000 feet	Cold bend mandrel diameter
	Number of pairs	Number of strands	Size of strands	Minimum	Maximum	Minimum	Maximum	Minimum	Nominal
			AWG	Inch.	Inches	Volts	Loop, ohms	Megohms	Inches
PBTM-5	5	7	30	0.062	0.590	1000	0.46	100	8
PBTM-15	15	7	30	.062	.800	1000	.46	100	10
PBTM-30	30	7	30	.062	1.030	1000	.46	100	13

"

Page 63, paragraph 3.47: Delete and substitute:

"3.47 Type DLT cable. -

CONSTRUCTION

First. A stranded steel strength member conforming to Specification MIL-C-5424, 7 x 19 strand, 5/32 inch in diameter, not preformed insulated with 0.046 inch minimum wall (see 3.7.3) of 60 percent natural rubber in accordance with the applicable requirements of Publication ASTM D27-54T to a nominal diameter of 0.262 inch. The insulation shall fill all outer interstices of the steel strand. This wall may be applied in one or two layers and shall be properly vulcanized so that the completed cable meets all test requirements.
Second. A stranded conductor of coated copper, size 1-1/2(41) (see 3.10), covered with a cotton separator (see 3.11.12.2) and insulated with 0.031 inch minimum wall (see 3.7.3 and prior to cabling) of 30 percent natural rubber in accordance with Publication ASTM D27-54T. Circuit identification (see 3.9.7).
Third. Four insulated conductors shall be cabled in a close helix about the insulated strength member with a lay of approximately 0.625 inch. The insulated conductors shall be partially embedded in the relatively soft partially vulcanized insulation of the strength member.
Fourth. A minimum 0.031 inch belt (see 3.7.3) of butyl rubber (see 3.11.5) filling the outer interstices over the insulated conductor assembly.

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Fifth. Cotton reinforcement (see 3. 11. 12. 3).
Sixth. A polychloroprene sheath (see 3. 11. 17) having a minimum thickness (see 3. 7. 3) of 0. 078 inch bonded to the underlying butyl belt. The assembled components shall be so molded and vulcanized that the completed cable shall be one compact mass from which the individual conductors shall be readily separable without damage to the insulation. The overall diameter shall not exceed 0. 710 inch. Manufacturer's identification (see 3. 8) shall include molded marking (see 3. 8. 4) at intervals not greater than 25 feet, showing the cable type letter designation (type DLT), the manufacturer's name or trade mark, and the year of manufacture.

LENGTH REQUIREMENT

Type DLT cable shall be delivered in nominal lengths of not less than 600 feet and not more than 630 feet. Except where authorized by the contracting agency, not more than 30 percent of the total footage of each lot offered for inspection may be accepted in random lengths of not less than 400 feet and not more than 420 feet or in random lengths of not less than 200 feet and not more than 210 feet. Combinations thereof of random lengths will be permitted. Other lengths are not suitable for the purpose.

TEST REQUIREMENTS

Preproduction testing - Type DLT cable is not required to have approval as a qualified product (see 3. 1 and 6. 3) prior to award. In lieu thereof, the manufacturer shall submit via the Government inspector to the Material Laboratory, New York Naval Shipyard, 200 feet of completed cable for preproduction testing. All of the following tests including those listed under inspection testing shall be met satisfactorily at the Material Laboratory prior to design approval by the inspector for the manufacturer to enter into production under the contract or order.
Water and air leakage (see 4. 8. 13) - No leakage at 550 pounds per square inch.
Flexing endurance (see 4. 8. 14) - 1000 bending cycles only on 2 inch mandrel at room temperature.
Breaking strength (see 4. 8. 28) - 2500 pounds minimum.
Endurance test - 500 cycles without conductor breakage or without electrical failures between conductors or conductor to strength member. Cycle to consist of drawing specimen under alternate loads of 490 and 180 pounds tension back and forth once through 90 degrees around a 3 inch diameter pulley with a total travel of 18 inches in one direction.
Inspection testing - The following tests shall be performed at place of manufacture under supervision of the inspector.
Lot acceptance inspection (see 4. 4).
Physical and aging tests on conductor insulation and sheath (see 4. 5. 2).
Flammability (see 4. 8. 10).
Dielectric strength - 5 minutes, 2000 volts r. m. s. (min.) between conductors and from conductor to ground (strength member and water bath).
Insulation resistance - 200 megohms per 1000 feet (min.).
Conductor resistance - 17. 0 ohms per 1000 feet at 25°C. (max.)."

Page 65, paragraph 3. 49, after tabulation: Add "The nominal length or unit ordering length for type MRI-D-1 is a 600-foot coil".

Page 66, paragraph 3. 50: Delete and substitute:

"3. 50 Type T3P cable. -

CONSTRUCTION

First. Conductors and ground wires of coated copper wires (see 3. 10 and following table).
Second. A wall of butyl rubber (see 3. 11. 5) on insulated conductors.
Third. (a) A color-coded tape (see 3. 11. 15) or an inner braid (see 3. 11. 16) on each insulated conductor.
(b) A green inner braid (see 3. 11. 16) on each uninsulated ground wire.

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Fourth. Three insulated conductors twisted together with a left-hand lay not greater than twenty times the pitch diameter. One ground wire shall be placed in each of the three valleys with sufficient fillers (see 3. 11. 11) to obtain a firm, well-rounded assembly. A compound-filled tape (see 3. 11. 15) over the conductor assembly.

Fifth. A polychloroprene sheath (see 3. 11. 17). Reinforcement (see 3. 11. 12. 3) shall be applied under the sheath or approximately near the middle of the sheath thickness so as not to show as an overall surface marking.

TEST REQUIREMENTS

Flammability (see 4. 8. 10).
Cold bend (see 4. 8. 9). No damage when bent two turns at -30°C. around mandrel with diameter twice that of specimen.
Electrical tests (see following table).

Type	Conductors		Insulation thickness	Sheath thickness	Overall diameter	Dielectric strength	Conductor resistance (d. c.), 1000 feet at 25°C.	Insulation resistance per 1000 feet
	Number	Stranding	Minimum	Minimum	Maximum	Minimum	Maximum	Minimum
			Inch	Inch	Inches	Volts	Ohm	Megohms
T3P-42								
Insulated	3	7x7/0.0292	0.125	0.172	1.67	11,500	0.273	500
Uninsulated	3	7x7/ .0164	0	-----	----	0	.874	0
T3P-400								
Insulated	3	37x7/ .0393	0.141	0.266	3.20	13,000	.0284	250
Uninsulated	3	19x7/ .0316	0	-----	----	0	.0863	0

3.51 Workmanship. - The workmanship shall be first class in every respect and uniform throughout each lot of cable offered for delivery."

Page 69, paragraph 4. 5. 2. 1. 1: Add as items (g) and (h):

"(g) Induced voltage (see 4. 8. 29).
"(h) Watertightness (see Appendix I)."

Page 70, paragraph 4. 6. 2: Add "Thermoplastic flow (see 4. 8. 32)."

Page 71, paragraph 4. 7. 2: Add "Thermoplastic flow (see 4. 8. 32)."

Page 73, paragraph 4. 8. 1. 2. 1. 5: Delete and substitute:

"4. 8. 1. 2. 1. 5 Number of specimens. - Samples of insulation shall be removed from each conductor of cables comprising four or less conductors, and from five of the conductors selected from cables having more than four conductors. Five specimens of insulation shall be prepared from each of the conductors so selected. Five specimens shall be prepared from each sample of sheath compound selected for test. If

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the values taken from the first three specimens of each sample tested are all satisfactory, tests on the other two specimens need not be conducted. If one or more of the first three specimens fail, the remaining two specimens shall be tested and the physical property of the sample shall be considered to be the median of the results obtained from the five specimens. The median value of the aged and unaged property shall be used in calculating the deterioration."

Page 75, paragraph 4.8.8.1: Delete and substitute:

"4.8.8.1 Specimen. - The specimen shall be an 18-inch section of completed cable."

Page 75, paragraph 4.8.9.3, line 9: Delete "cold bend test, or" and substitute "cold bend test, of".

Page 76, paragraph 4.8.9.3.2, line 2: Delete types MHFF (see 3.36) and MHFA (see 3.23)" and substitute "type MHFF (see 3.36)".

Page 77, table XIV, column 1: Delete reference to "FCOTP-4, SRI and SRIB", and after "FHOF", add "SSF, T3P, TSP, DLT, JAS, DSS, TSS, FSS and MSS".

Page 79, table XVI: Delete columns 2, 3 and 4.

Page 80, paragraph 4.8.12.1, lines 3 and 4: Delete "and type MHFA (see 3.23) which are designated as control cables" and substitute "which is designated as control cable".

Pages 81 and 82, paragraph 4.8.14.3.2, tabulation, column 1: Line 1, add "TTRS-6"; line 2, add "TTRS-8"; line 3, add "TTRS-10, 12", and line 4, add "TTRS-16".

Page 85, paragraph 4.8.19.6.3, line 1: Delete "types SRIB and" and substitute "type".

Page 87, table XVIII, column 3, line 3: Delete "Thin-walled cables (see 1.2.3)".

Page 90, paragraph 4.8.29.1, line 1: Delete "MCCS-5 and MCOS-6" and substitute "MCOS-5, MCOS-6 and MSS-6".

Page 90, paragraph 4.8.29.3.1, line 3: Delete "between conductors at the open end" and substitute "across the 150 ohm resistor at either end."

Page 90, paragraph 4.8.29.3.2, line 1: After "MCOS-6", add "and MSS-6" and in line 2, delete "between conductors at the open end" and substitute "across the 150 ohm resistor at either end".

Page 90, paragraph 4.8.29.4: Delete.

Page 91, paragraph 4.8.31, line 4: Delete "18" and substitute "6".

Page 91: Add as paragraph 4.8.32:

"4.8.32 Thermoplastic flow test. - A specimen of the completed cable with the ends prepared for electrical testing shall be wound for two turns around a mandrel having a diameter of 10 to 12 times the overall cable diameter. The coiled specimens shall be placed in an air oven at 150°C. with a 60 cycle potential of 120 volts between each conductor and shield and 120 volts between each conductor and its mate."

Page 91, paragraph 5.1.1.1, tabulation: Delete and substitute:

"MRI-T-2-1/2	MMOP-5	SHOF-3	DSS
SHFS-14	DBSP	DHOF-4	TSS
DCOP-2	TBSP	THOF-4	FSS
TCOP-2	FBSP	FHOF-3	MSS"

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Page 93, paragraph 5.3.4.2, tabulation: Delete and substitute:

Year of manufacture		Identifying color
"1955	1960	White
1956	1961	Red
1957	1962	Green
1958	1963	Orange
1959	1964	Blue"

Page 95, table XIX: Delete reference to types "SHFA, SHFP, DHFA, THFA, FHFA, MHFA, FCSF-220, SRI, SRIB, DSS, TSS, FSS and MSS"

Page 95, table XIX: In column 4, under "TTOP", add "TSP and PBTM"; in column 6, opposite FHOF133, delete "1000" and substitute "500"; in column 7, under "DHOF", add "JAS" and under "THOF", add "T3P"

Page 95, table XIX: Add the following types:

Designation		Feet
Types	Sizes	Nominal
"SSF	300	[4]1100
DLT	4	[5]615"

Page 95, table XIX: Add the following footnotes:

"[4]See 3.12.
[5]See 3.47."

Page 96, table XX, between "Coil" and "Reel" insert:

Type of package	Nominal length	Standard lengths	Random lengths	Remnant lengths		Scrap lengths
				Price reduction		
		No price reduction	No price reduction	5 percent	10 percent	Not acceptable
	Feet	Feet	Feet	Feet	Feet	Feet
"Coil	600	660 to 540	539 to 180	179 to 120	119 to 60	59 to 0"

Page 96, table XX, column 6, line 2: Delete "159 to 80" and substitute "99 to 50".

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Page 97: Delete reference to types "CHFA, FCOTP-4, FHFA, MHFA, SHFA, SHFP, SRI, SRIB and THFA" and add the following types:

Type	Individual specification sheet
"DLT	3.47
JAS	3.45
PBTM	3.46
SSF	3.12
T3P	3.50"

Pages 101 and 102, appendix I: Delete and substitute:

"APPENDIX I - WATERTIGHTNESS

10. Purpose. - The manufacturer shall endeavor to fill all types of watertight cable (see 3.6) so that no water can be forced into or through the cable core, and so that the electrical characteristics of the cable are not affected, by exposure of the open ends to water under pressure. The quality of the watertight feature shall be judged by the total amount of water which leaks from the open free end of the specimen, when tested as specified herein.

20. Sampling. - One 5-foot specimen from every manufactured length (see 3.2) of watertight cable shall be tested and the leakage found shall represent the watertight quality of the entire length from which the specimen is cut. After a specimen representing the manufactured length has been tested and the manufactured length is cut subsequently in two or more pieces for delivery under different contracts or to different destinations, specimens shall not be cut from the several pieces for additional watertightness tests unless specifically requested by the manufacturer. The value of specimen lengths consumed in retesting shall be borne by the manufacturer and shall not be included in the contract or order. In the event of retests, the last test only shall determine the relative acceptability of the material represented and the results of previous tests shall be disregarded.

30. Procedure. - One end of the 5-foot specimen shall be subjected to a continuous period of 6 hours to tap water under 25 pounds per square inch pressure. The method employed for applying the water pressure shall not constrict nor permit expansion of the specimen within the gland. Since different packing methods will influence the results, and because all plants must be judged on the same basis, all tests shall be made with a low-melting point (not exceeding 190°F.) metallic alloy packing material. No other method shall be employed without prior approval in writing by the bureau or agency concerned.

40. Permissible leakage. - The following tabulation shows several degrees of water leakage for each type and size of watertight cable. These values are based on an analysis of inspection tests data, and the values in column 1 are set at a low level which may be difficult to meet, unless manufacturers continue to cooperate in this development by discovering and introducing new refinements to improve watertight quality. Watertight quality is improved continuously in this manner. Eventually, zero leakage will be specified. Failure of a reasonable number of specimens to conform to the column 1 values is expected at the present stage of the development and is not considered as evidence of substandard material, but as evidence of the need for tighter quality control and care in processing.

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50. Limits. - The following tabulation, and the explanatory notes thereto, constitute a part of this specification:

All cable types with conductor sizes	Total water leakage during 6 hour test at 25 p.s.i.	
	Column 1	Column 2
	No price reduction	10 percent price reduction
	Cubic inches	Cubic inches
9 and smaller	1.0 or less	1.1 or more
14 to 40 incl.	5.0 or less	5.1 or more
50 to 100 incl.	10.0 or less	10.1 or more
125 to 300 incl.	20.0 or less	20.1 or more
350 and larger	30.0 or less	30.1 or more

Note 1. - The water leakage reported for any specimen will determine whether the cable footage represented by that specimen comes within the limits of column 1 or column 2.

Note 2. - Cable footage represented by specimens which do not exceed the leakage permitted in column 1 is acceptable at the contract price.

Note 3. - Cable footage represented by specimens which exceed the leakage permitted in column 1 is acceptable under column 2 at a 10 percent reduction in the contract price for that footage, provided the watertight sealing compounds are found to be present as specified in 3.6.

60. Reports. - The results of all tests and retests made at the plant under lot acceptance inspection shall be reported, together with the footage represented by each specimen, to the Bureau of Ships by the Government inspector. The results of all tests made at the Material Laboratory, New York Naval Shipyard, shall be reported to the cognizant Government inspector with a copy to the Bureau of Ships."

Preparing activity:
Navy - Bureau of Ships
(Project 6145-0027Sh)

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